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To cite this article: Rajavath Narayana Naik *et al* 2024 *J. Phys.: Conf. Ser.* **2765** 012003

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Impact of Exponential Heat Source and Thermal Radiation on the flow of Hybrid Nanofluid across a Bi-Directional Stretching Surface with Activation Energy

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Abstract. The study focuses on how activation energy and exponential heat source affects the radiative motion of a hybrid nanofluid ($EG + ZnO + TiO_2$) over a bidirectional elongating sheet. Convective boundary condition is assumed. The bvp4c, an inherent function in MATLAB, is used to unravel the altered system deduced from those equations which describe the current work in the two scenarios, i.e., binary ($EG + ZnO + TiO_2$) and mono ($EG + ZnO$) nanofluids. Added a little quantitative discussion regarding engineering parameters including Sherwood number. The most notable findings from this study are mentioned here. Enhanced volume fraction of ZnO lessens the fluid velocity and increasing heat source parameter step-up the fluid temperature. It was noticed that with a raise in magnetic field and the volume fraction of ZnO lessens the friction factor. The rate of mass transfer is lowered with the enhanced activation energy parameter and the same enhances as the reaction rate parameter is amplified. The increment quantities in the Sherwood number against the reaction rate parameter (Γ) are 0.35151 ($EG + ZnO + TiO_2$) and 0.35125 ($EG + ZnO$), when Γ is set to $0 \leq \Gamma \leq 0.9$. Furthermore, the decrement rates in mass transfer rate against the activation parameter (E_n) are observed as 0.02681 ($EG + ZnO + TiO_2$) and 0.02687 ($EG + ZnO$), when E_n is set to $0 \leq E_n \leq 2.5$. It is detected that there is an amelioration in the heat transmission rate with the raise in Biot number (Bi). It has been noticed that, when Bi fixed to be $0 \leq Bi \leq 0.8$, Nusselt number is lowered by 0.736332 ($EG + ZnO + TiO_2$) and 0.621826 ($EG + ZnO$). Our findings are checked against previous findings for validity. A reasonable compromise has been uncovered.

Keywords: Hybrid nanofluid, Bidirectional stretching sheet, Activation energy, Heat source, bvp4c.



1. Introduction

Nanofluids becomes a novel and intriguing class of heat transfer fluids which can be used in place of more conventional options. Heat exchangers, Fuel cells and pharmaceutical procedures are few examples for them. Saba et al. [1] scrutinized the radiative nanofluid flow (Water + SWCNT/MWCNT) by a curled elongating surface together with heat sink/source. Prabavathi et al. [2] considered the same nanofluid and inspected the chemically reactive and radiative Maxwell fluid flow by a vertical cone using FEM (finite element method). Upsurges in the Maxwell parameter is observed to result in an increase in the fluid's temperature. Basha et al. [3] elucidated the radiative nanofluid (water+EG/Diamond and water+EG/SWCNH) flow via a plate/wedge. Saranya and Al-Mdallal [4] carried out a study on various shapes of nanoparticles in the radiative MHD nanofluid (Silicon oil + Al_2O_3) flow by a revolving disk. The heat transmission rate of nanoparticles with a blade form is measured to be significantly higher than that of other shapes. Newly, different researchers [5-7] assumed distinct geometries and analyzed different nanofluid flows including $Ag - Water$ nanofluid. Devi and Devi [8] discussed the features of heat transfer in a HNF ($Cu - Al_2O_3 / Water$) by an elongating sheet. They reported that the enhancement in the heat transmission rate is nearly 17.3% high equated to pure water for nano particle volume fraction. Bioconvective flow of HNF (Water + Copper + Silica) by a revolving disc was modelled mathematically by Jayadevamurthy et al. [9], who utilized a mix of RKF-45 and shooting procedures to unravel the problem. Mabood et al. [10] did an irreversibility analysis on the radiative HNF ($Cu - Al_2O_3 / H_2O$) flow by a melting surface and noticed the increase in heat transmission as the radiation parameter was increased. Khashiie et al. [11] analyzed the same HNF flow by a shrinking/elongating sheet with various conditions including slip with `bvp4c` function (MATLAB in-built). Acharya and Mabood [12] numerically investigated a hybrid nanofluid (HNF) motion across a slippery bended surface subject to a nonlinear thermal radiation. Freshly, many studies [13- 22] assumed various geometries and examined numerous HNF flows under various conditions, including quadratic velocity. With the increase in nanoparticle volume fraction, it is noticed a 5.5 percent increase in friction factor (2 percent to 4 percent).

Many different industries, including electricity production and solar energy, rely on insights from the heat transmission study of boundary layer flow when radiation is taken into account. Salomatov and Puzyrev [23] provided an analysis for the convective flow of a fluid by a flat surface with thermal radiation in two cases i.e., large and small assesses of radiation. Similar geometry was taken into account by Hossain and Takhar [24], who theoretically explored the radiation-mixed convective flow of a fluid. Researchers have shown that higher levels of radiation lead to a reduction in shear stress. The numerical investigation of the radiative Darcy-Forchheimer flow of an optically dense fluid by a wedge was reported by Al-Odat et al. [25] using the Keller box scheme. It has been noticed that the temperature of the fluid decreases when the heat sink is deployed. Abdel-Wahed [26] modelled the radiative MHD nanofluid flow through a proceeding surface and observed that the magnetic field impact lessens the heat transmission rate. Khan et al. [27] carried out research on the Williamson fluid flow through an elongated surface with the aid of Lie group procedure. Observations show that when the Weissenberg number rises, so does the momentum boundary layer thickness. Sheremet et al. [28] numerically analyzed the radiative convective flow of a nanofluid ($Al_2O_3 + H_2O$) within a tended square cavity. They emphasized that the lean angle of cavity could be a better regulator factor for fluid flow. With the magnetic field effect, on the same fluid, Sheikholeslami et al. [29] studied the radiative convective Darcy flow inside a wavy surface. Later, different researchers [30-40] discussed distinct radiative nanofluid flows through different geometries under various conditions.

On the radiative Hybrid nanofluid flow, ($EG + ZnO + TiO_2$) via a bidirectional elongated sheet, no works were not yet proposed to reveal the impact of activation energy and diverse heat source

parameters. SLR (Slope of Linear Regression) and graphs are used to discuss engineering characteristics of relevance, such as skin friction coefficient. The current results are also justified by comparing them with the past results.

2. Formulation

In this study, the radiative hybrid nanofluid motion across a bidirectional elongating surface with activation energy and diverse heat source parameter was investigated. As a base fluid, Ethylene glycol is considered, ZnO and TiO_2 are taken to be nanoparticles for hybrid nanofluid considered. Table-1 displays the numerical values of their thermophysical parameters. Magnetic field with intensity B_0 is enforced in the z - direction (observe Fig. 1). We assumed $U_w(x) = a_1x$, the elongating velocity of the sheet in x -direction, and it is $V_w(y) = a_2y$ in y -direction.

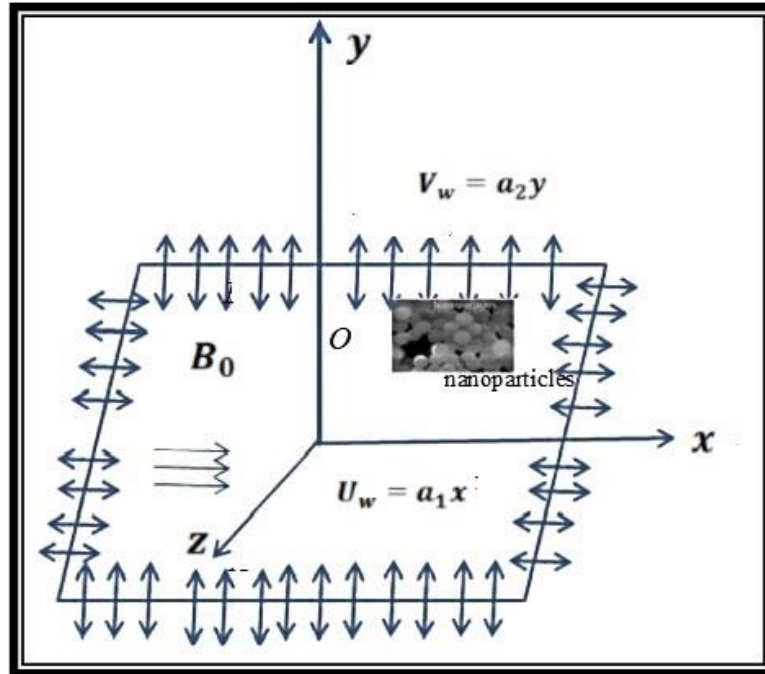


Figure 1. Flow geometry (Gupta and Gupta [41])

Under these premises, governing equations and boundary conditions are presented as (Gupta and Gupta [41], Al-Kouz et al. [42] and Irfan et al. [43]):

$$\frac{\partial}{\partial x}(u) + \frac{\partial}{\partial y}(v) + \frac{\partial}{\partial z}(w) = 0 \quad (1)$$

$$\rho_{hmf} \left(u \frac{\partial}{\partial x}(u) + v \frac{\partial}{\partial y}(u) + w \frac{\partial}{\partial z}(u) \right) = \mu_{hmf} \frac{\partial}{\partial z} \left(\frac{\partial}{\partial z}(u) \right) - \sigma_{hmf} B_0^2 u - u \frac{\mu_{hmf}}{\kappa} \quad (2)$$

$$\rho_{hmf} \left(u \frac{\partial}{\partial x}(v) + v \frac{\partial}{\partial y}(v) + w \frac{\partial}{\partial z}(v) \right) = \mu_{hmf} \frac{\partial}{\partial z} \left(\frac{\partial}{\partial z}(v) \right) - v \sigma_{hmf} B_0^2 - \frac{\mu_{hmf}}{\kappa} v \quad (3)$$

$$\begin{aligned} (\rho C_p)_{hmf} \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) &= \left(k_{hmf} + \frac{16}{3} \frac{\sigma^* T_\infty^3}{k^*} \right) \frac{\partial^2 T}{\partial z^2} \\ + q_e (T_w - T_\infty) \exp \left(-n \sqrt{\frac{a}{\nu}} z \right) &+ q_t (T - T_\infty) \end{aligned} \quad (4)$$

$$u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} + w \frac{\partial C}{\partial z} = D_m \frac{\partial^2 C}{\partial z^2} - k_0^2 \left(\frac{T}{T_\infty} \right)^m \exp \left(-\frac{E^*}{k_1 T} \right) (C - C_\infty) \quad (5)$$

$$\begin{aligned} \text{at } z = 0: u = U_w, v = V_w, w = 0, -k_{hmf} \frac{\partial T}{\partial z} &= h_f (T_w - T), C = C_w \\ \text{as } z \rightarrow \infty: u \rightarrow 0, v \rightarrow 0, T \rightarrow T_\infty, C \rightarrow C_\infty & \end{aligned} \quad (6)$$

Gupta and Gupta [41] proposed the usage of the similarity variables listed below in the transformation of governing equations:

$$\begin{aligned} \eta = z \sqrt{\frac{U_w}{\nu x}}, u = a_1 x f', w = -\sqrt{a_1 \nu} (f(\eta) + g(\eta)), v = a_1 y g', \\ \theta(\eta) = \frac{T - T_\infty}{T_w - T_\infty}, \Phi(\eta) = \frac{C - C_\infty}{C_w - C_\infty}. \end{aligned} \quad (7)$$

2.1. Hybrid nanofluid (HNF) thermophysical characteristics (Hanif et al. [44])

$$\begin{aligned} (\rho C_p)_{hmf} &= \left[(1 - \phi_1) (\rho C_p)_f + \phi_1 (\rho C_p)_{s_1} \right] (1 - \phi_2) + (\rho C_p)_{s_2} \phi_2, \\ \mu_{hmf} &= \frac{\mu_f}{(1 - \phi_1)^{2.5} (1 - \phi_2)^{2.5}}, \rho_{hmf} = \left[(1 - \phi_1) \rho_f + \phi_1 \rho_{s_1} \right] (1 - \phi_2) + \rho_{s_2} \phi_2, \\ k_{hmf} &= k_{nf} \times \frac{k_{s_2} + 2k_{nf} - 2k_{nf} \phi_2 + 2k_{s_2} \phi_2}{k_{s_2} + 2k_{nf} + k_{nf} \phi_2 - k_{s_2} \phi_2}, k_{nf} = k_f \times \frac{k_{s_1} + 2k_f - 2k_f \phi_1 + 2k_{s_1} \phi_1}{k_{s_1} + 2k_f + k_f \phi_1 - k_{s_1} \phi_1}, \\ \sigma_{hmf} &= \sigma_{nf} \times \frac{\sigma_{s_2} + 2\sigma_{nf} - 2\sigma_{nf} \phi_2 + 2\sigma_{s_2} \phi_2}{\sigma_{s_2} + 2\sigma_{nf} + \sigma_{nf} \phi_2 - \sigma_{s_2} \phi_2}, \sigma_{nf} = \sigma_f \times \frac{\sigma_{s_1} + 2\sigma_f - 2\sigma_f \phi_1 + 2\sigma_{s_1} \phi_1}{\sigma_{s_1} + 2\sigma_f + \sigma_f \phi_1 - \sigma_{s_1} \phi_1}. \end{aligned}$$

Next, (7) makes (1) true, rewrites (2-5) and (6) as:

$$\frac{1}{B_1 B_2} f''' + f f'' + g f'' - f'^2 - \left(\frac{B_3}{B_2} M + \frac{\delta}{B_1 B_2} \right) f' = 0 \quad (8)$$

$$\frac{1}{B_1 B_2} g''' + f g'' + g g'' - g'^2 - \left(\frac{B_3}{B_2} M + \frac{\delta}{B_1 B_2} \right) g' = 0 \quad (9)$$

$$\left(\frac{B_4}{B_5} \frac{1}{\text{Pr}} + \frac{1}{B_5} \frac{Ra}{\text{Pr}} \right) \theta'' + (f + g) \theta' + \frac{Q_T}{B_5} \theta + \frac{1}{B_5} Q_E \exp(-m\eta) = 0 \quad (10)$$

$$\frac{1}{Sc} \Phi'' + (f + g) \Phi' - \Gamma (1 + \Lambda \theta)^m \exp\left(-\frac{E_n}{1 + \Lambda \theta}\right) \Phi = 0 \quad (11)$$

$$\left. \begin{aligned} \text{at } \eta = 0: f(\eta) = g(\eta) = 0, f'(\eta) = 1, g'(\eta) = \lambda, \theta'(\eta) = -Bi(1 - \theta(\eta)), \Phi(\eta) = 1 \\ \text{as } \eta \rightarrow \infty: f'(\eta) \rightarrow 0, g'(\eta) \rightarrow 0, \theta(\eta) \rightarrow 0, \Phi(\eta) \rightarrow 0 \end{aligned} \right\} \quad (12)$$

Note that

$$\left. \begin{aligned} B_1 &= (1 - \phi_1)^{2.5} (1 - \phi_2)^{2.5}, B_4 = \frac{k_2 + 2B_{41}k_f - 2\phi_2(B_{41}k_f - k_2)}{k_2 + 2B_{41}k_f + \phi_2(B_{41}k_f - k_2)}, \\ B_3 &= \frac{\sigma_2 + 2B_{31}\sigma_f - 2\phi_2(B_{31}\sigma_f - \sigma_2)}{\sigma_2 + 2B_{31}\sigma_f + \phi_2(B_{31}\sigma_f - \sigma_2)}, B_{41} = \frac{k_1 + 2k_f - 2\phi_1(k_f - k_1)}{k_1 + 2k_f + \phi_1(k_f - k_1)}, \\ B_{31} &= \frac{\sigma_1 + 2\sigma_f - 2\phi_1(\sigma_f - \sigma_1)}{\sigma_1 + 2\sigma_f + \phi_1(\sigma_f - \sigma_1)}, B_2 = (1 - \phi_2) \left[(1 - \phi_1) + \phi_1 \frac{\rho_1}{\rho_f} \right] + \phi_2 \frac{\rho_2}{\rho_f}, \\ B_5 &= (1 - \phi_2) \left[(1 - \phi_1) + \phi_1 \frac{(\rho C_p)_1}{(\rho C_p)_f} \right] + \phi_2 \frac{(\rho C_p)_2}{(\rho C_p)_f}. \end{aligned} \right\}$$

and

$$\left. \begin{aligned} \delta &= \frac{\nu_f}{a_1 \kappa}, \Lambda = \frac{T_w - T_\infty}{T_\infty}, Ra = \frac{16}{3} \frac{\sigma^* T_\infty^3}{k_f k^*}, \Gamma = \frac{k_0^2}{a_1}, Q_E = \frac{q_e}{a_1 (\rho C_p)_f}, \lambda = \frac{a_2}{a_1}, \\ Q_T &= \frac{q_t}{a_1 (\rho C_p)_f}, M = \frac{\sigma_f B_0^2}{\rho_f a_1}, E_n = \frac{E^*}{k_1 T_\infty}, Sc = \frac{\nu}{D_m}, Bi = \frac{h_f}{k_f \sqrt{a_1 \nu}}. \end{aligned} \right\}.$$

In this work, we assume the condition (without losing generality) $(f + g)(0) = 0$ as $f(0) = g(0) = 0$.

Table 1 Base fluid and nanomaterial thermophysical property values (Jamshed and Aziz [45])

S. No.	Properties	EG (<i>f</i>)	ZnO (<i>s</i> ₁)	TiO ₂ (<i>s</i> ₂)
1	σ	0.0000055	0.01	2380000
2	k	0.252	13	8.953
3	ρ	1114	5660	4250
4	C_p	2415	495.2	686.2

Skin friction coefficient, transfer rates (heat, mass) are defined as:

$$Cf_x = \frac{\tau_{zx}}{\frac{1}{2}\rho_f u_w^2} \Big|_{z=0}, Cf_y = \frac{\tau_{zy}}{\frac{1}{2}\rho_f v_w^2} \Big|_{z=0}, Nu_x = \frac{xq_w}{k_f(T_w - T_\infty)} \Big|_{z=0}, Sh_x = \frac{xs_w}{D_m(C_w - C_\infty)} \Big|_{z=0} \quad (13)$$

where

$$\tau_{zx} = \mu_{mf} \frac{\partial u}{\partial z}, \tau_{zy} = \mu_{mf} \frac{\partial v}{\partial z}, q_w = -\left(k_{mf} + \frac{16}{3} \frac{\sigma^* T_\infty^3}{k^*}\right) \frac{\partial T}{\partial z}, s_w = -D_m \frac{\partial C}{\partial z}.$$

Terms in (13) can be recast using (7) as:

$$\begin{aligned} (\text{Re}_x)^{0.5} Cf_x &= \frac{2}{B_1} f''(0), (\text{Re}_y)^{0.5} Cf_y = \frac{2}{B_1} \frac{1}{\sqrt{\lambda}} g''(0), \\ (\text{Re}_x)^{-0.5} Nu_x &= -B_4 (1 + Ra) \theta'(0), (\text{Re}_x)^{-0.5} Sh_x = -\Phi'(0) \end{aligned} \quad (14)$$

3. Numerical Procedure

For obtaining the solution of the system (8 - 11) subject to the conditions (12), we employed bvp4c, an inherent function in MATLAB. The flowchart for the bvp4c numerical technique is presented in Fig.2.

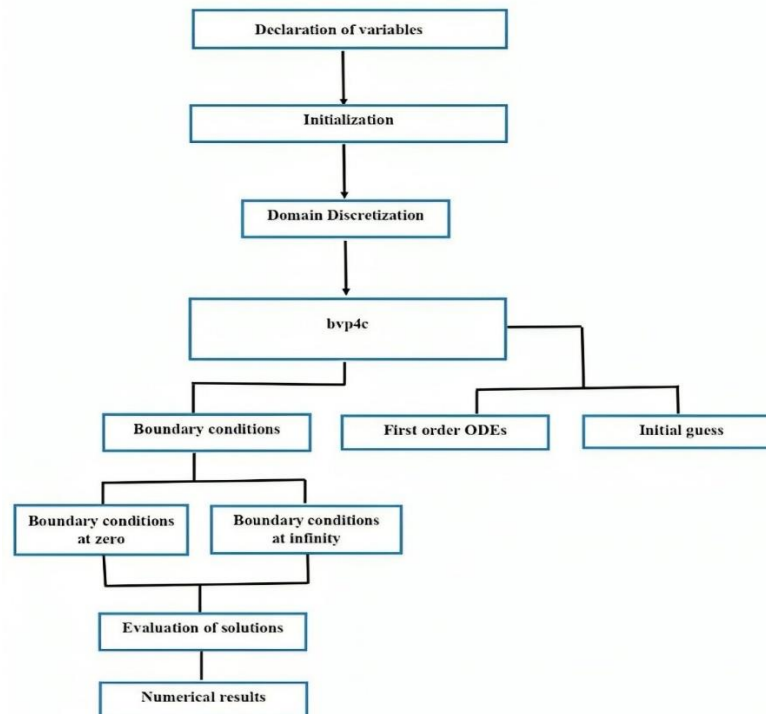


Figure 2. Schema chart for bvp4c (Faiz et al. [46])

4. Validation

Table 2 compares our findings to those of other studies under special situations and reveals a very close agreement.

Table 2 Verification of current findings by referencing previously obtained data for $-\theta'(0)$ when $\phi_1 = \phi_2 = M = \delta = Ra = Q_T = Q_E = \Gamma = 0, \lambda = 1$.

Pr	Khan and Pop [47]	Gupta and Gupta [41]	Current results
0.07	0.066	0.06632019	0.066319
0.20	0.169	0.16831182	0.168314
0.70	0.454	0.45423652	0.454228
2.0	0.911	0.91148210	0.911480

5. Discussion of the outcomes

Results are displayed in the form of graphs for two instances i.e., $EG + ZnO + TiO_2$ and $EG + ZnO$. Figure 3 depicts streamlines representing the current motion of the fluid.

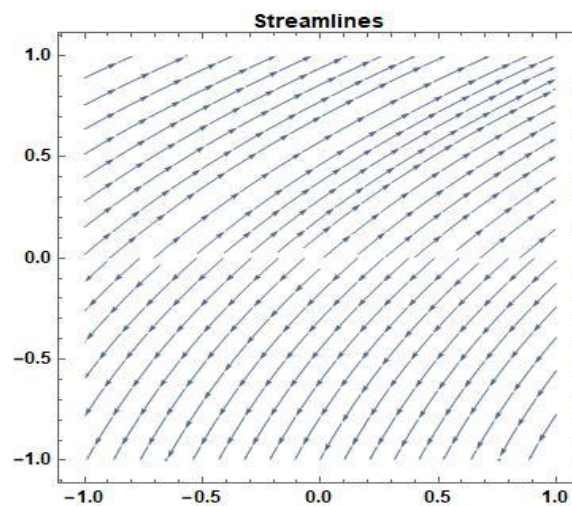


Figure 3. Streamlines for the present flow study

When there is a magnetic field, Lorentz forces are typically generated by the motion of an electrically conducting liquid. Because of the slowing effect of Lorentz forces, fluid velocities drop as magnetic field is increased (Figs. 4 - 5). As the nanoparticle volume percentage rises, the fluid becomes more viscous and becomes more difficult to move through. Therefore, when ϕ_1 grows, velocity fall (Figs. 6 - 7)

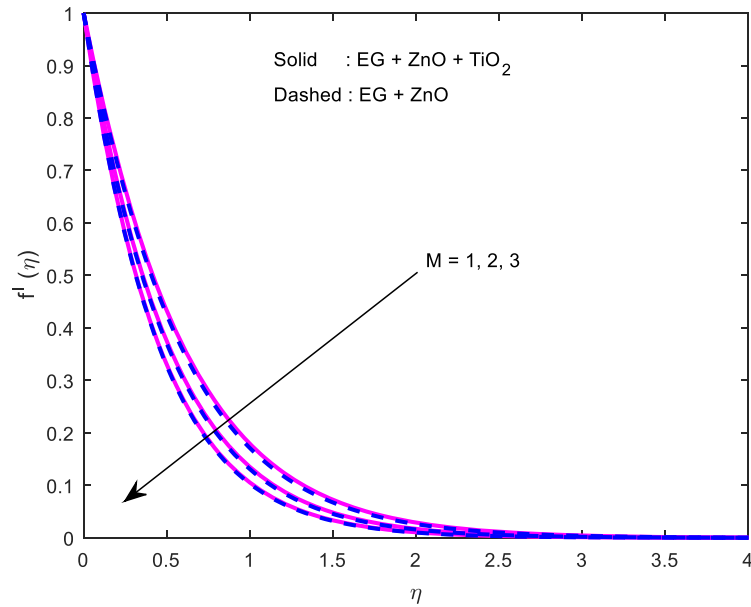


Figure 4. Behavior of $f'(\eta)$ when M is applied

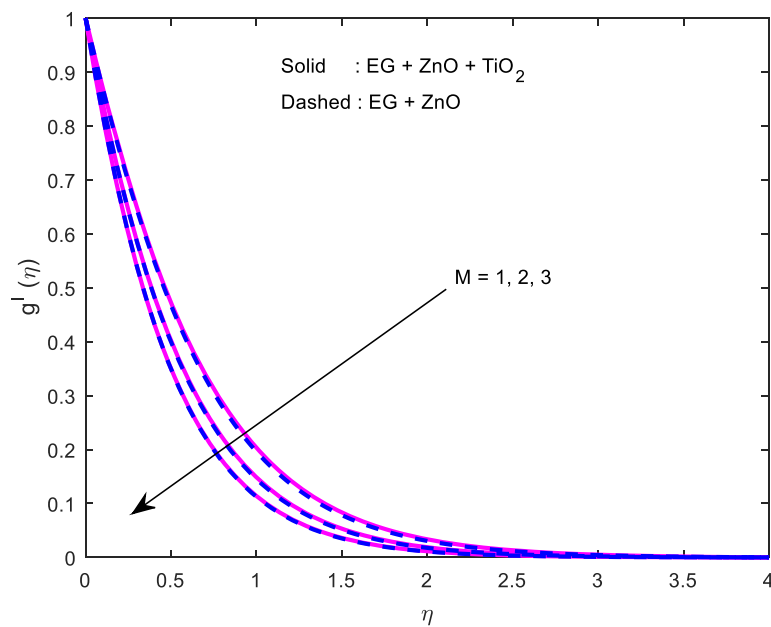
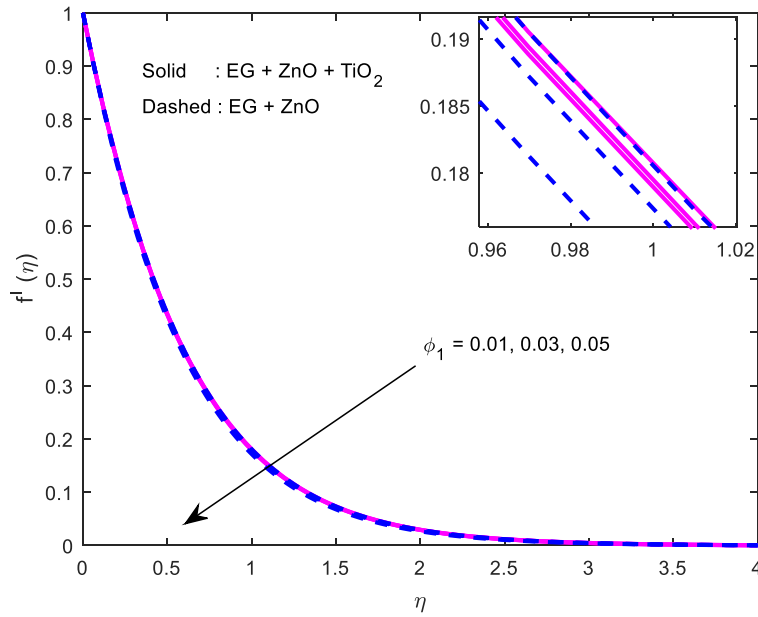
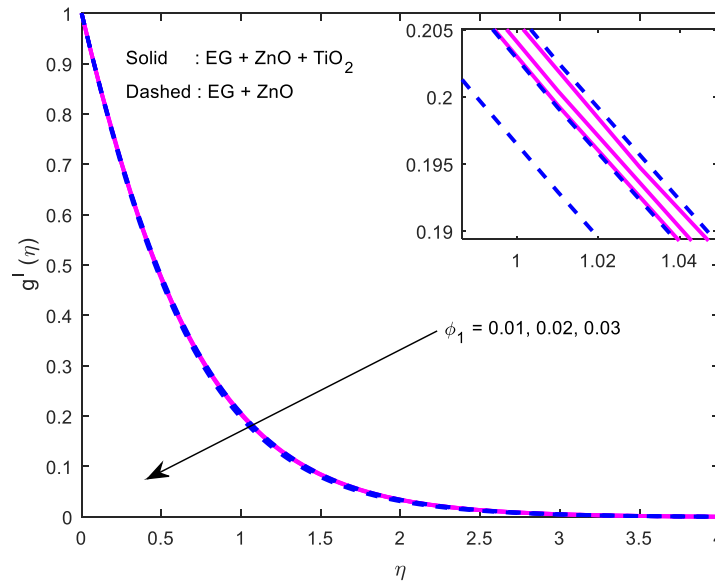


Figure 5. Result of M on $g'(\eta)$

Figure 6. Result of ϕ_1 on $f'(\eta)$ Figure 7. Behavior of $g'(\eta)$ when ϕ_1 is applied

Under the influence of thermal radiation on the flow, amount of heat is transferred into the fluid. If that value is enhanced, there is a huge amount of transfer of heat energy is noticed from source to fluid region. As a result, as illustrated in Fig.8, due to the enhancement in temperature causes to the absorbance of heat energy in fluid particles. More fluid particle collisions take place when the nanoparticles' volume percentage rises. In turn, this causes the fluid's temperature to increase (see Fig. 9). From Fig. 10, it was noticed that, there is an enhancement in fluid temperature due to the increment in Q_T . Fig. 11 shows the acceleration of temperature profile with a rise in Q_E in both scenarios. The ESHS mechanism (Heat Source connects to Exponential space) is the reason for a rise in fluid temperature while enhancing the heat source parameter, which leads to transfer of more energy into the

liquid. The rate of heat transmission via convection in a fluid rises in proportion to its Biot number. As a reason, temperature of the fluid is augmented (as depicted in Fig. 12).

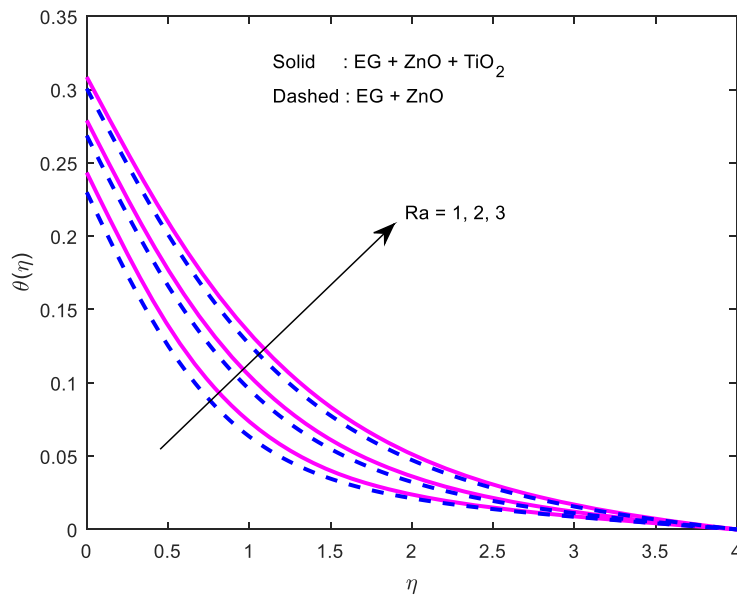


Figure 8. Profile of $\theta(\eta)$ due to impact of Ra

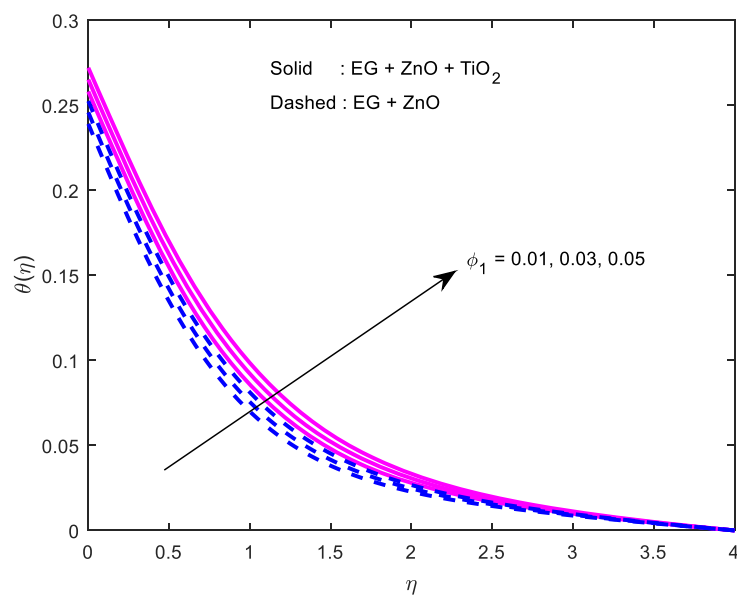
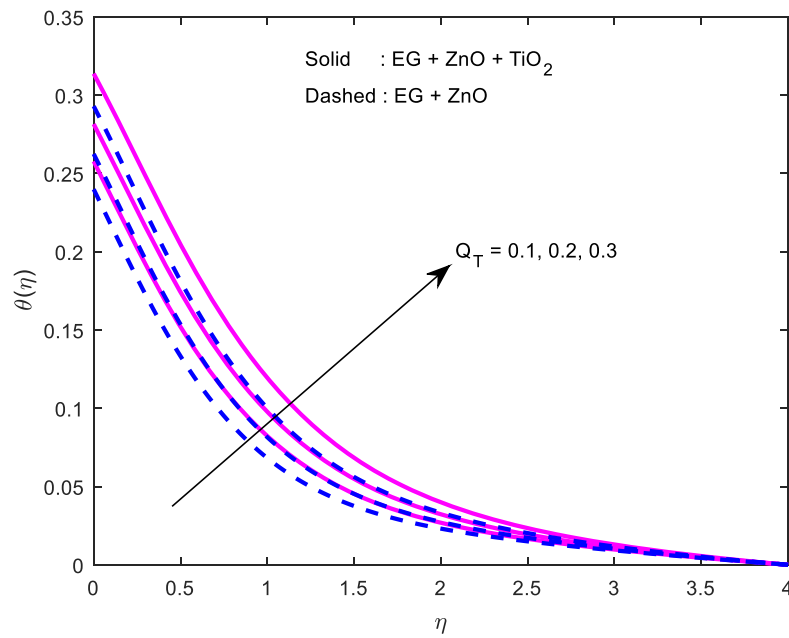
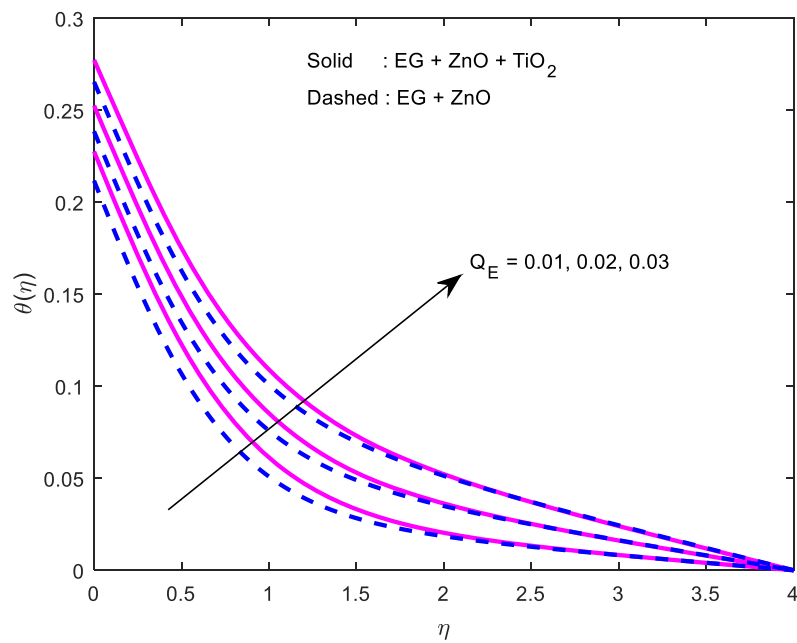


Figure 9. Profile of $\theta(\eta)$ due to impact of ϕ_1

Figure 10. Profile of Q_T due to impact of $\theta(\eta)$ Figure 11. Profile of Q_E due to impact of $\theta(\eta)$

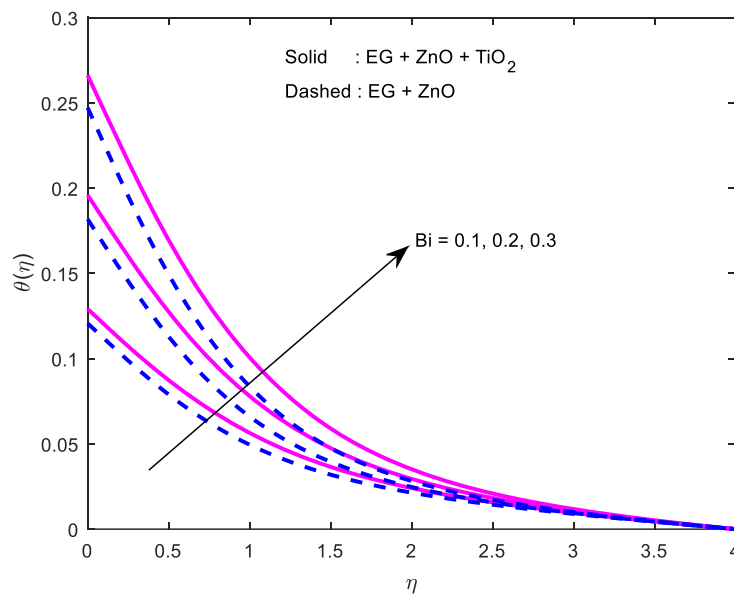


Figure 12. Temperature profile when Bi is applied

By tracking the decline in the threshold energy caused by a rise in activation energy, one can determine the average kinetic energy of a fluid. From this, we can infer a drop in average kinetic energy. If there is no diffusion, there is a high fluid concentration (see Fig.13). As the reaction rate parameter rises, the fluid concentration seems to decrease (see Fig.14). Lower molecular diffusivity is caused by a greater temperature difference. Accordingly, the fluid concentration drops with increasing temperature difference parameter (see Fig. 15).

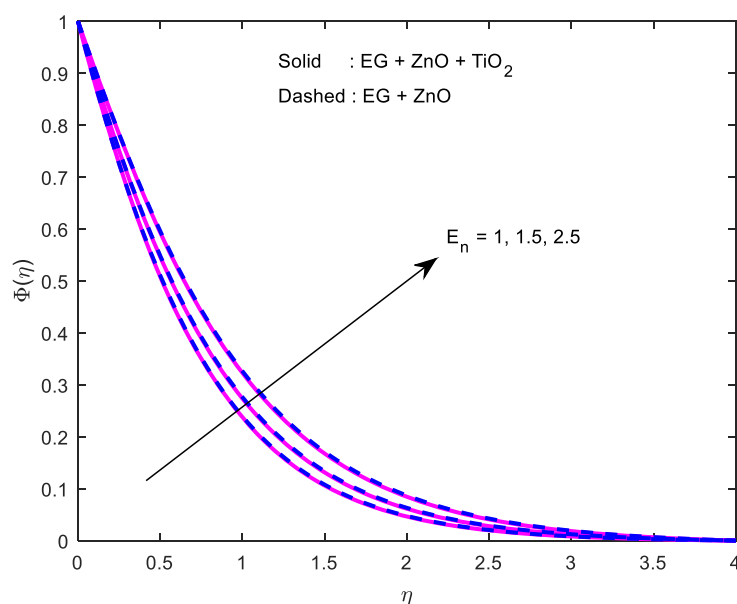
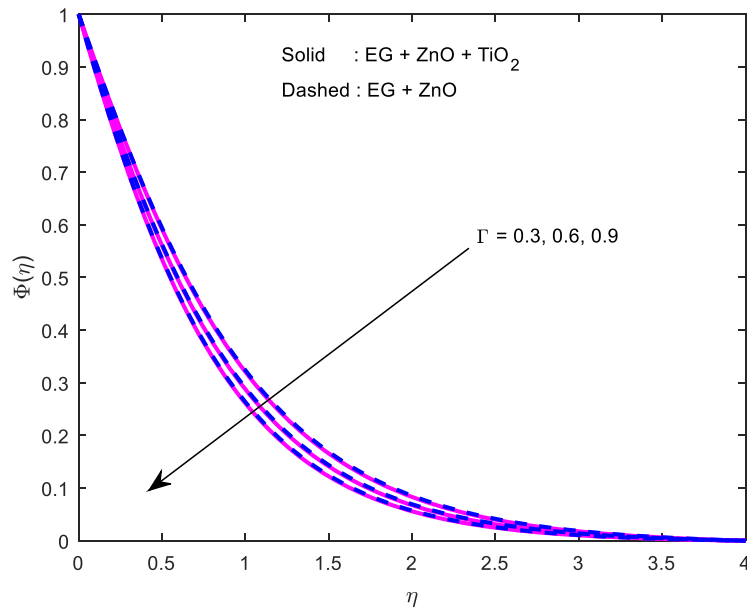
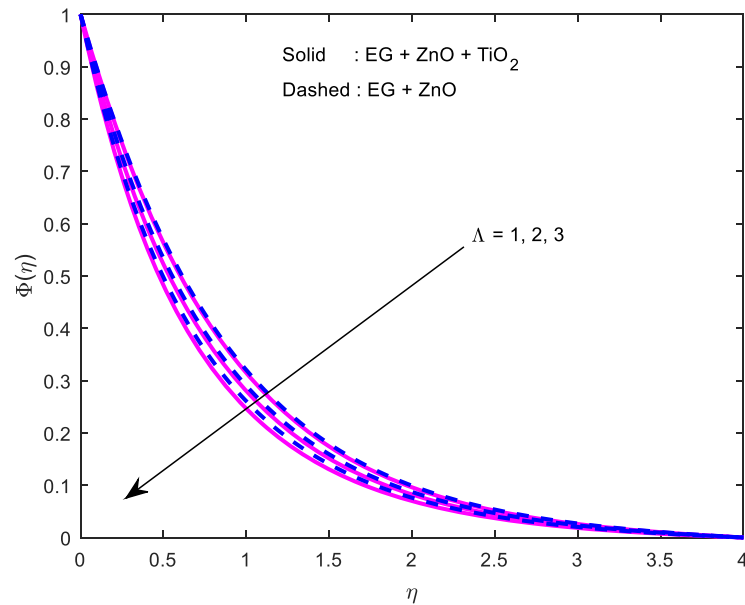


Figure 13. Profile of $\Phi(\eta)$ due to impact of E_n

Figure 14. Profile of $\Phi(\eta)$ due to impact of Γ Figure 15. Profile of $\Phi(\eta)$ due to impact of Λ

From Figs. 16-17, it is clear that both M and ϕ_1 are useful to reduce the friction factor in both situations namely, Hybrid Nanofluid ($EG + ZnO + TiO_2$) and Mono Nanofluid ($EG + ZnO$). It can be shown in Fig. 18 that when Q_T upsurges, the heat transmission rate declines. Reduction quantities in the heat transfer rate against Q_T are noticed as 0.15677 ($EG + ZnO + TiO_2$), 0.11564 (MNF). Therefore, it is worthy to mention that reduction rate is higher in case of $EG + ZnO$ compared to $EG + ZnO + TiO_2$. Fig. 19 exhibits the fact that there is an amelioration in the Nusselt number with the raise in Bi . Increment quantity is detected as 0.736332 ($EG + ZnO + TiO_2$) and 0.621826 ($EG + ZnO$).

).From Figs. 20-22, it is discovered that the activation energy parameter diminishes the Sherwood number and Γ, Λ escalate the same. The increment quantities against Γ, Λ are 0.35151 ($EG + ZnO + TiO_2$), 0.35125 ($EG + ZnO$), 0.04911 ($EG + ZnO + TiO_2$) and 0.04300 ($EG + ZnO$) respectively. Furthermore, the decrement in mass transfer rate against the activation parameter is observed as 0.02681 ($EG + ZnO + TiO_2$) and 0.02687 ($EG + ZnO$).

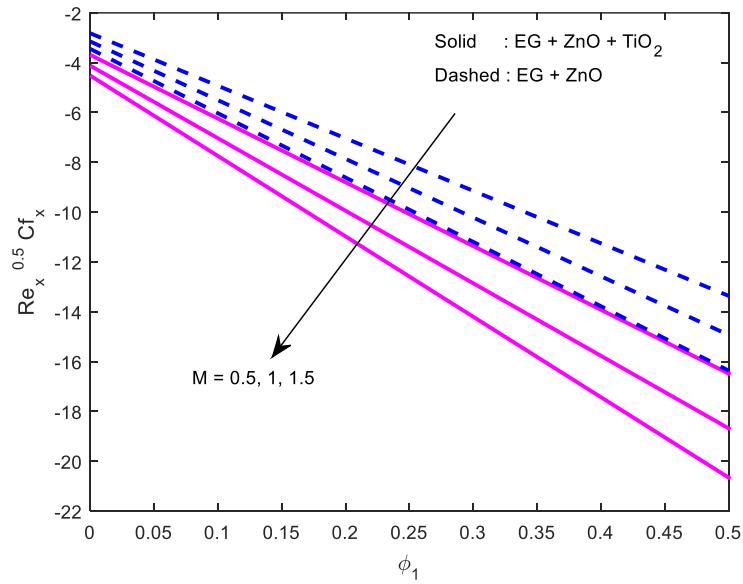


Figure 16. Behavior of skin friction coefficient when M is applied

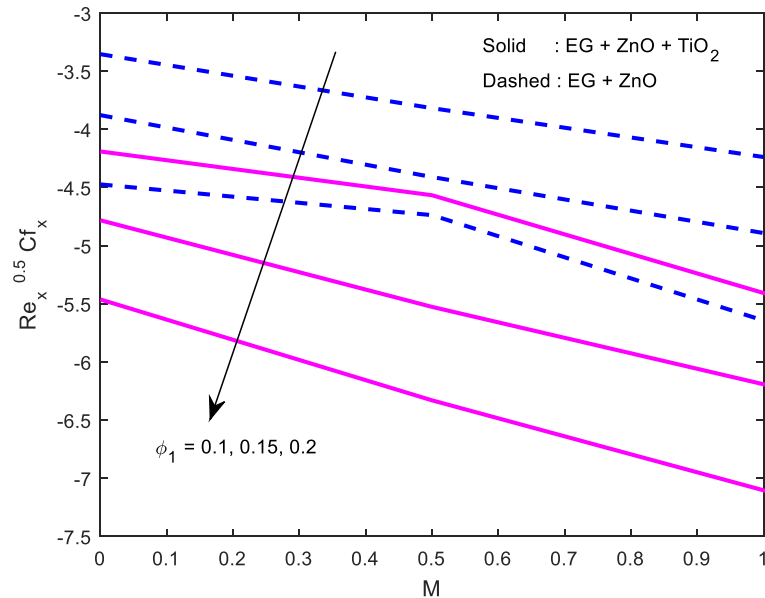


Figure 17. Behavior of skin friction coefficient when ϕ_1 is applied

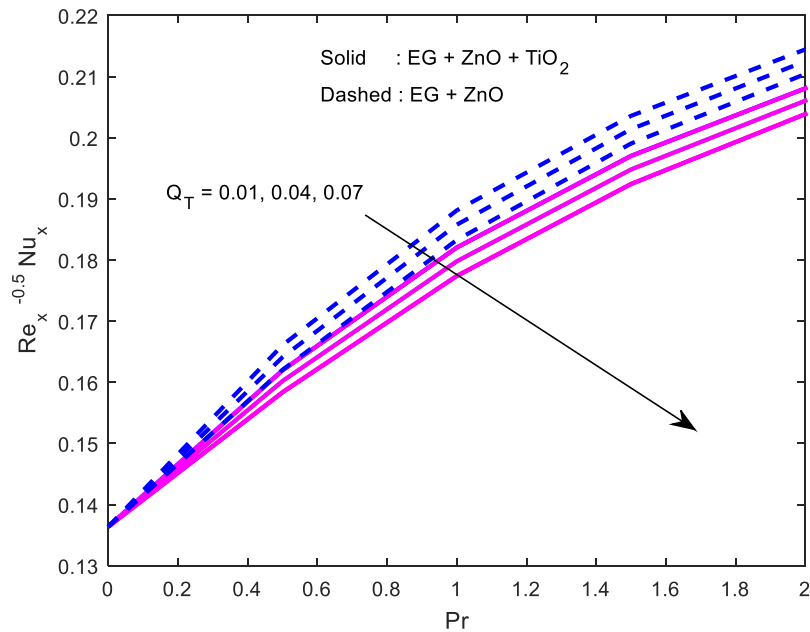


Figure 18. Behavior of heat transmission rate when Q_T is applied

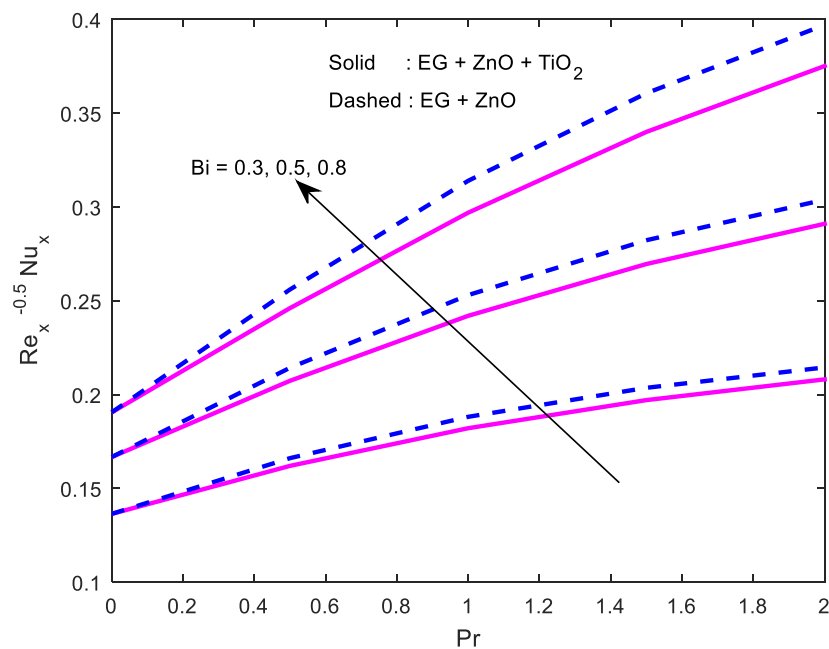


Figure 19. Behavior of heat transmission rate when Bi is applied

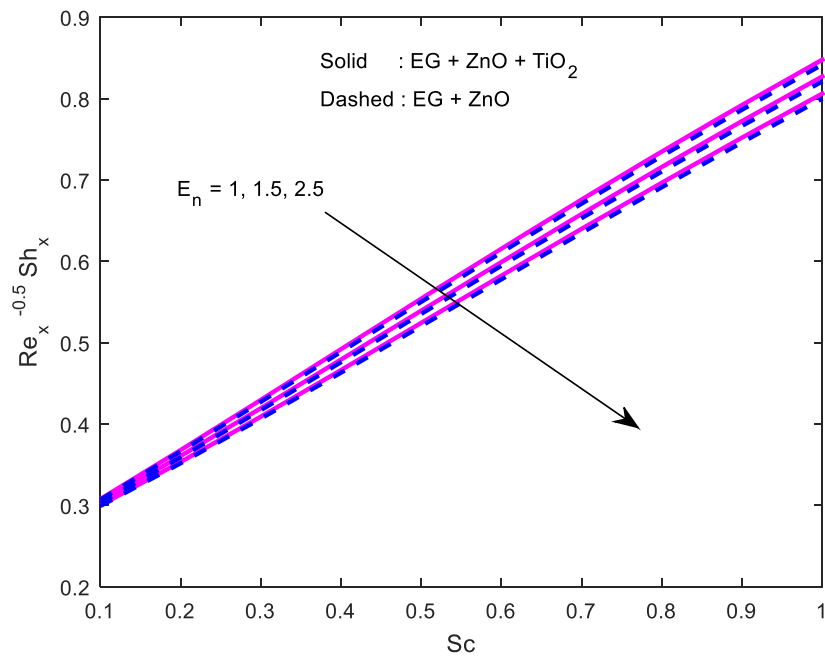


Figure 20. Behavior of mass transmission rate when E_n is applied

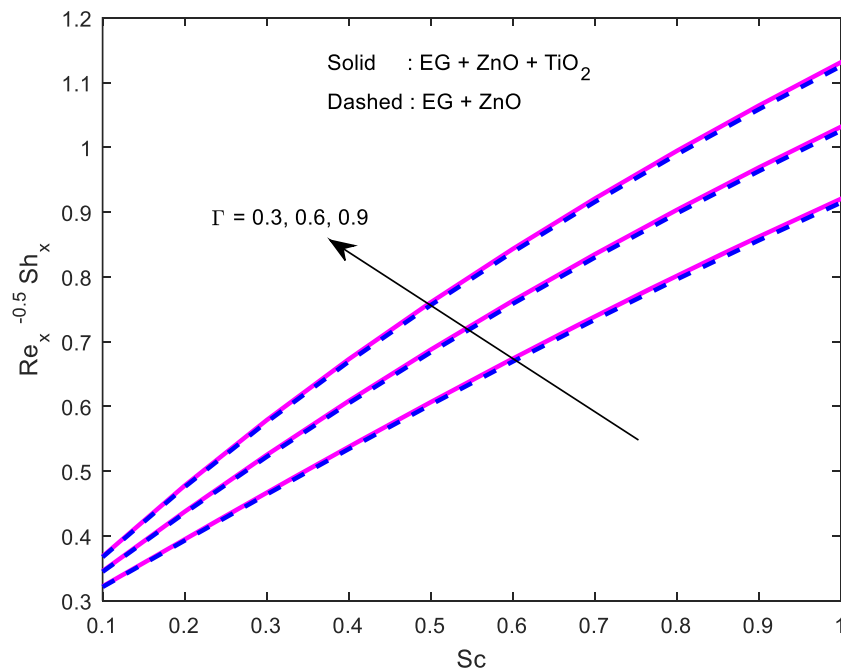


Figure 21. Behavior of mass transmission rate when Γ is applied

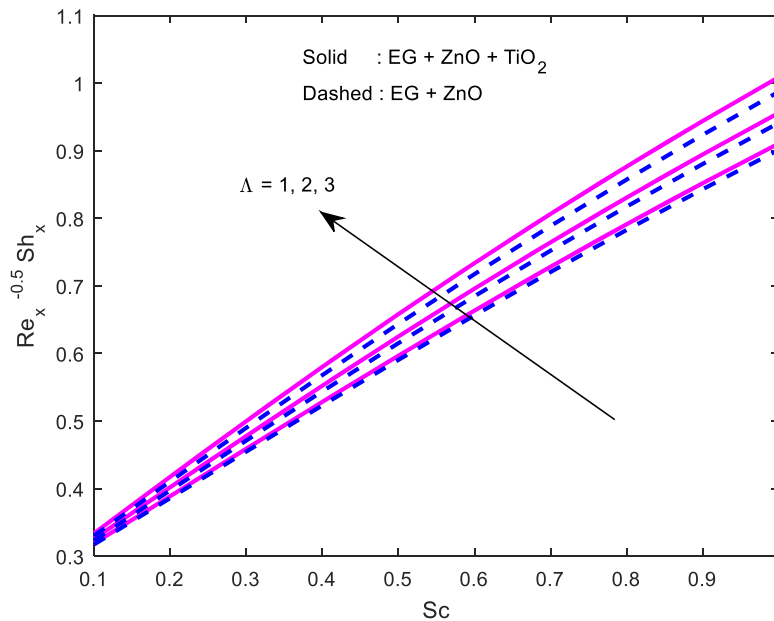


Figure 22. Behavior of mass transmission rate when Λ is applied

6. Conclusion

In this study, radiative hybrid nanofluid ($EG + ZnO + TiO_2$) flow by a bidirectional elongating sheet with different heat source parameters and activation energy is analyzed theoretically. Convective boundary condition is assumed. The results of this investigation were obtained using a pre-existing function in MATLAB called bvp4c. Added a little quantitative discussion regarding engineering parameters inclusive of Sherwood number. The concluding remarks of this study are briefly summarized below.

- Enhanced volume fraction of ZnO lessens the fluid velocity and increasing the heat source parameter step-up the fluid temperature.
- It was observed that, due to the occurrence of magnetic field as well as the volume fraction of ZnO is beneficial in lowering the skin friction coefficient in both scenarios.
- It is discovered that the activation energy parameter lowers the Sherwood number. The decrement rates in the Sherwood number against the activation parameter (E_n) are observed as 0.02681 ($EG + ZnO + TiO_2$) and 0.02687 ($EG + ZnO$), when E_n is set to $0 \leq E_n \leq 2.5$.
- It is detected that Γ and Λ are helpful to enhances the mass transfer rate. Sherwood number is increased with a rate of 0.35151 ($EG + ZnO + TiO_2$) and 0.35125 ($EG + ZnO$), when Γ is set to $0 \leq \Gamma \leq 0.9$.
- There is an amelioration in the heat transmission rate with the raise in Bi . It was perceived that, when Bi is set to $0 \leq Bi \leq 0.8$, Nusselt number is reduced to 0.736332 ($EG + ZnO + TiO_2$) and 0.621826 ($EG + ZnO$).

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